

# Dynamic Lagrangian Frequency–Time (DLFT) in admittance notation for Frequency-Based Substructuring

Companion note for `pyhbm` – class `FBS_DLFT`

May 23, 2026

## Abstract

This note derives the Dynamic Lagrangian Frequency–Time (DLFT) harmonic-balance method of Vadcard *et al.* (2022) directly in *admittance* (receptance/FRF) notation and embeds it in a Lagrange-multiplier Frequency-Based Substructuring (LM-FBS) assembly that uses signed Boolean matrices  $\mathbf{B}_c, \mathbf{B}_e$  on a block-diagonal substructure admittance  $\mathbf{Y} = \text{blkdiag}(\mathbf{Y}^1, \mathbf{Y}^2, \dots)$ . The closed-form coupling of Soleimani & Aage (2025, Eq. 24) is recovered as the two-substructure special case. We give the residual, the *full* analytic Jacobian  $\partial \mathbf{r} / \partial \tilde{\mathbf{x}}_r$  and  $\partial \mathbf{r} / \partial \omega$  (including the conditional contact term and the  $\partial \mathbf{Y} / \partial \omega$  chain rule), state exactly where the DFT/IDFT enter, and answer the algorithmic question: *the primary Newton unknown is the multiharmonic vector of relative-displacement Fourier coefficients  $\tilde{\mathbf{x}}_r$ , not the contact forces  $\tilde{\boldsymbol{\lambda}}$* . The derivation is the specification for the `FBS_DLFT` class and the predictor–corrector continuation loop in `pyhbm`.

## Contents

### 1 Notation and harmonic-balance preliminaries

Consider a linear elastic substructure with mass, damping and stiffness matrices  $\mathbf{M}, \mathbf{C}, \mathbf{K} \in \mathbb{R}^{d \times d}$  driven at the fundamental angular frequency  $\omega$ . With the (real) Fourier ansatz for a  $T = 2\pi/\omega$  periodic displacement,

$$\mathbf{u}(t) = \text{Re} \sum_{n \in \mathcal{H}} \hat{\mathbf{u}}_n e^{in\omega t}, \quad \mathcal{H} = \{0, 1, \dots, N_H\}, \quad N_h = |\mathcal{H}|, \quad (1)$$

each harmonic  $n$  obeys the algebraic balance

$$\mathbf{Z}_n(\omega) \hat{\mathbf{u}}_n = \hat{\mathbf{f}}_n, \quad \mathbf{Z}_n(\omega) = \mathbf{K} - (n\omega)^2 \mathbf{M} + in\omega \mathbf{C} \in \mathbb{C}^{d \times d}, \quad (2)$$

where  $\mathbf{Z}_n$  is the *dynamic stiffness* (impedance) and  $\mathbf{Y}_n(\omega) = \mathbf{Z}_n(\omega)^{-1}$  the *admittance* (receptance/FRF) of harmonic  $n$ . Stacking harmonics,  $\tilde{\mathbf{x}}_r, \tilde{\mathbf{f}} \in \mathbb{C}^{N_h d}$ , the block-diagonal multiharmonic operators are

$$\mathbf{Z}(\omega) = \text{blkdiag}_{n \in \mathcal{H}} \mathbf{Z}_n(\omega), \quad \mathbf{Y}(\omega) = \mathbf{Z}(\omega)^{-1} = \text{blkdiag}_{n \in \mathcal{H}} \mathbf{Y}_n(\omega). \quad (3)$$

For a nonlinear system with internal force  $\mathbf{f}_{\text{nl}}$  the multiharmonic balance residual reads, in impedance form (Vadcard Eq. 23; Krack §2),

$$\mathcal{H}(\hat{\mathbf{u}}, \omega) = \mathbf{Z}(\omega) \hat{\mathbf{u}} + \tilde{\mathbf{f}}_{\text{nl}}(\hat{\mathbf{u}}) - \tilde{\mathbf{f}}_{\text{ext}} = \mathbf{0}. \quad (4)$$

**DFT/IDFT operators.** Nonlinear and contact forces are evaluated in the time domain on  $N_t$  equispaced samples  $\tau_i = 2\pi i/N_t$ ,  $i = 0, \dots, N_t - 1$  ( $N_t$  over-samples the band  $\mathcal{H}$ ). Let

$$\mathbf{\Gamma} : \mathbb{C}^{N_h d} \rightarrow \mathbb{R}^{N_t d} \text{ (synthesis / IDFT)}, \quad \mathbf{\Gamma}^+ : \mathbb{R}^{N_t d} \rightarrow \mathbb{C}^{N_h d} \text{ (analysis / DFT)}, \quad (5)$$

so that a band-limited real signal  $\mathbf{s}(t_i) = \mathbf{\Gamma} \hat{\mathbf{s}}$  and  $\hat{\mathbf{s}} = \mathbf{\Gamma}^+ \mathbf{s}$ , with  $\mathbf{\Gamma}^+ \mathbf{\Gamma} = \mathbf{I}$  on the retained band. Acting on the real/imaginary split  $\hat{\mathbf{s}} = \hat{\mathbf{s}}_R + i\hat{\mathbf{s}}_I$ , the (real) entries of  $\mathbf{\Gamma}, \mathbf{\Gamma}^+$  for one DOF are

$$\mathbf{\Gamma}_{i,n}^{(R)} = \frac{c_n}{N_t} \cos(n\tau_i), \quad \mathbf{\Gamma}_{i,n}^{(I)} = -\frac{c_n}{N_t} \sin(n\tau_i), \quad \mathbf{\Gamma}_{n,i}^{+(R)} = \cos(n\tau_i), \quad \mathbf{\Gamma}_{n,i}^{+(I)} = -\sin(n\tau_i), \quad (6)$$

with  $c_0 = 1$ ,  $c_{n>0} = 2$  (the `numpy.rffft/irfft` convention used in `pyhbm`). The distinction “ $\mathbf{\Gamma}$  acts on the *real* pair  $(\hat{\mathbf{s}}_R, \hat{\mathbf{s}}_I)$ ” is essential for the Jacobian (§??).

## 2 LM-FBS assembly and condensation to the relative interface DOF

### 2.1 Dual (Lagrange-multiplier) assembly

Let substructures  $s = 1, \dots, N_s$  have admittances  $\mathbf{Y}^s(\omega)$ . Concatenate the global, *uncoupled* quantities

$$\mathbf{u} = \text{col}(\mathbf{u}^1, \dots, \mathbf{u}^{N_s}), \quad \mathbf{f} = \text{col}(\mathbf{f}^1, \dots, \mathbf{f}^{N_s}), \quad \mathbf{Y} = \text{blkdiag}(\mathbf{Y}^1, \dots, \mathbf{Y}^{N_s}), \quad (7)$$

(each understood per harmonic, or block-diagonal over  $\mathcal{H}$  as in (??)). Following van der Seijs (Eqs. 3.20, 3.25), coupling is imposed through a *compatibility* Boolean  $\mathbf{B}_c$  and an *equilibrium* Boolean  $\mathbf{B}_e$ : the connecting forces are  $\mathbf{g} = -\mathbf{B}_e^\top \boldsymbol{\lambda}$  and compatibility reads  $\mathbf{B}_c \mathbf{u} = \mathbf{0}$ . The dual system is

$$\mathbf{u} = \mathbf{Y}(\mathbf{f} - \mathbf{B}_e^\top \boldsymbol{\lambda}), \quad \mathbf{B}_c \mathbf{u} = \mathbf{0}. \quad (8)$$

For *collocated*, *conforming* interfaces (the case implemented here)  $\mathbf{B}_c = \mathbf{B}_e =: \mathbf{B}$  is the signed Boolean with  $\pm 1$  on each connected DOF pair. Eliminating  $\boldsymbol{\lambda}$  for a *rigid* joint gives the classical LM-FBS one-liner (van der Seijs Eqs. 3.21–3.23)

$$\boldsymbol{\lambda} = (\mathbf{B}_c \mathbf{Y} \mathbf{B}_e^\top)^{-1} \mathbf{B}_c \mathbf{Y} \mathbf{f}, \quad \tilde{\mathbf{Y}} = \left[ \mathbf{I} - \mathbf{Y} \mathbf{B}_e^\top (\mathbf{B}_c \mathbf{Y} \mathbf{B}_e^\top)^{-1} \mathbf{B}_c \right] \mathbf{Y}. \quad (9)$$

### 2.2 Nonlinear interface: condensation onto the relative DOF

With a *nonlinear* interface the rigid constraint  $\mathbf{B}_c \mathbf{u} = \mathbf{0}$  is replaced by an interface constitutive law: the multiplier  $\tilde{\boldsymbol{\lambda}}$  becomes the (unknown) contact force that depends on the relative interface motion. Define the multiharmonic *relative interface displacement*

$$\tilde{\mathbf{x}}_r := \mathbf{B}_c \hat{\mathbf{u}} \in \mathbb{C}^{N_h n_{\text{int}}}, \quad (10)$$

( $n_{\text{int}}$  interface DOF pairs). Substituting the first line of (??) into (??),

$$\tilde{\mathbf{x}}_r = \mathbf{B}_c \mathbf{Y} \mathbf{f} - \mathbf{B}_c \mathbf{Y} \mathbf{B}_e^\top \tilde{\boldsymbol{\lambda}} = \mathbf{f}_{\text{adm}} - \mathbf{Y}_r \tilde{\boldsymbol{\lambda}}, \quad (11)$$

where we name the *interface admittance* and the *admittance (free) excitation response*

$$\boxed{\mathbf{Y}_r := \mathbf{B}_c \mathbf{Y} \mathbf{B}_e^\top} \quad (= \mathbf{Y}^1 + \mathbf{Y}^2 \text{ collocated, 2 subs.}), \quad \boxed{\mathbf{f}_{\text{adm}} := \mathbf{B}_c \mathbf{Y} \tilde{\mathbf{f}}_{\text{ext}}}. \quad (12)$$

Equation (??) rearranges to the **admittance FBS residual**

$$\boxed{\mathbf{r}(\tilde{\mathbf{x}}_r) = \tilde{\mathbf{x}}_r + \mathbf{Y}_r \tilde{\boldsymbol{\lambda}} - \mathbf{f}_{\text{adm}} = \mathbf{0}} \quad (13)$$

which is the master equation of this note. It is the admittance counterpart of Vadcard’s impedance condensation  $\mathbf{Z}_r \tilde{\mathbf{x}}_r + \tilde{\boldsymbol{\lambda}} - \tilde{\mathbf{f}}_{\text{ex},r} = \mathbf{0}$  (his Eq. 37): multiply (??) from the left by  $\mathbf{Z}_r := \mathbf{Y}_r^{-1}$  and use  $\mathbf{Z}_r \mathbf{f}_{\text{adm}} = \tilde{\mathbf{f}}_{\text{ex},r}$  to recover it. Hence the impedance and admittance residuals share the same root; they differ only by the left factor  $\mathbf{Y}_r$ , i.e. by conditioning (see §??).

### 2.3 Recovery of Soleimani’s closed form (two substructures)

For  $N_s = 2$  collocated substructures,  $\mathbf{B}_c = \mathbf{B}_e = [\mathbf{I} \quad -\mathbf{I}]$  on the boundary DOFs,  $\mathbf{Y} = \text{blkdiag}(\mathbf{Y}^1, \mathbf{Y}^2)$ , so

$$\mathbf{Y}_r = \mathbf{B}\mathbf{Y}\mathbf{B}^\top = \mathbf{Y}_{bb}^1 + \mathbf{Y}_{bb}^2, \quad \mathbf{f}_{\text{adm}} = \mathbf{Y}_{be}^1 \hat{\mathbf{f}}_e^1 - \mathbf{Y}_{be}^2 \hat{\mathbf{f}}_e^2, \quad (14)$$

and (??) reads  $\tilde{\mathbf{x}}_r + (\mathbf{Y}_{bb}^1 + \mathbf{Y}_{bb}^2) \tilde{\boldsymbol{\lambda}} - (\mathbf{Y}_{be}^1 \hat{\mathbf{f}}_e^1 - \mathbf{Y}_{be}^2 \hat{\mathbf{f}}_e^2) = \mathbf{0}$ , i.e. exactly Soleimani & Aage Eq. (24) with  $\tilde{\boldsymbol{\lambda}} = \mathbf{f}_N$  (their receptance  $\mathbf{H} \equiv \mathbf{Y}$ ). The grounded/unilateral case (“one substructure rigid”) sets  $\mathbf{Y}^2 \rightarrow \mathbf{0}$ , giving  $\mathbf{Y}_r = \mathbf{Y}_{bb}^1$ ,  $\mathbf{f}_{\text{adm}} = \mathbf{Y}_{be}^1 \hat{\mathbf{f}}_e^1$  – the wall test case in `examples/fbs_dnft_unilateral_spring`.

## 3 Unilateral contact and the DLFT slaving of $\tilde{\boldsymbol{\lambda}}$

### 3.1 Constraints

The interface is a unilateral (compression-only) contact. With the gap (clearance) function and contact pressure  $\lambda(t)$  the Hertz–Signorini–Moreau / unilateral contact conditions (UCC) are

$$g(t) \geq 0, \quad \lambda(t) \geq 0, \quad g(t) \lambda(t) = 0. \quad (15)$$

We use the (implemented) convention of a wall at relative offset  $g_0$ : the scalar relative coordinate  $x_r(t) = (\boldsymbol{\Gamma} \tilde{\mathbf{x}}_r)(t)$  may not pass the wall, so

$$g(t) = g_0 - x_r(t), \quad \text{penetration } p(t) = x_r(t) - g_0 = -g(t). \quad (16)$$

(Vadcard writes  $g = x_r + g_0$  with the opposite sign of  $x_r$ ; (??) is the mirror convention and is physically equivalent for a wall at  $+g_0$ .)

### 3.2 Prediction step (admittance form)

The contact force is not an independent unknown; it is *slaved* to  $\tilde{\mathbf{x}}_r$  by a prediction–correction (this is what makes DLFT cheap – it does *not* double the unknowns with explicit multipliers). Isolating the multiplier in the equilibrium relation and adding a gap penalty with factor  $\varepsilon > 0$  (Vadcard Eq. 40), the prediction is

$$\tilde{\boldsymbol{\lambda}}_p(\tilde{\mathbf{x}}_r) = \mathbf{Z}_r(\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r) - \varepsilon \tilde{\mathbf{g}} = \underbrace{\mathbf{Z}_r(\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r)}_{\text{balancing force}} + \varepsilon(\tilde{\mathbf{x}}_r - \tilde{\mathbf{g}}_0), \quad \mathbf{Z}_r = \mathbf{Y}_r^{-1}. \quad (17)$$

The first term is the force that exactly satisfies the balance (??) at the current  $\tilde{\mathbf{x}}_r$  (note  $\mathbf{Z}_r(\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r) = \tilde{\mathbf{f}}_{\text{ex},r} - \mathbf{Z}_r \tilde{\mathbf{x}}_r$ , recovering Vadcard’s  $\tilde{\mathbf{f}}_{\text{ex},r} - \mathbf{Z}_r \tilde{\mathbf{x}}_r$ ). Forming  $\mathbf{Z}_r = \mathbf{Y}_r^{-1}$  from the substructure admittances is the one place the admittance assembly must *invert* the interface admittance  $\mathbf{Y}_r = \mathbf{B}_c \mathbf{Y} \mathbf{B}_e^\top$  – the same matrix whose conditioning van der Seijs (after Eq. 3.26) calls “highly defining for the success of the coupling.”

**Where the IDFT enters.** The UCC (??) are pointwise in time, so the prediction is taken to the time domain,

$$\lambda_p(t_i) = (\boldsymbol{\Gamma} \tilde{\boldsymbol{\lambda}}_p)_i = (\boldsymbol{\Gamma} \mathbf{Z}_r(\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r))_i + \varepsilon(x_r(t_i) - g_0). \quad (18)$$

Interpretation (Vadcard §2.4.2): where the gap is *violated* ( $p > 0$ ) the penalty drives  $\lambda_p$  strongly positive (contact detected); where the bodies *separate* ( $p < 0$ , i.e.  $g > 0$ ) it drives  $\lambda_p$  negative (to be removed below); in persistent contact ( $g \approx 0$ ) the penalty is negligible.

### 3.3 Correction step (UCC enforcement) and where the DFT enters

The predicted pressure violates  $\lambda \geq 0$  during separation. The correction enforces positivity *pointwise in time* (Vadcard Eq. 42),

$$\lambda(t_i) = \max(0, \lambda_p(t_i)) = \begin{cases} \lambda_p(t_i) & \lambda_p(t_i) > 0 \text{ (contact)} \\ 0 & \lambda_p(t_i) \leq 0 \text{ (separation)} \end{cases} \quad (19)$$

and is transformed back to the frequency domain by the *DFT*:

$$\tilde{\lambda}(\tilde{\mathbf{x}}_r) = \mathbf{\Gamma}^+ \{ \max(0, \lambda_p(t_i)) \}_{i=1 \dots N_t}. \quad (20)$$

Equations (??)–(??) define the map  $\tilde{\mathbf{x}}_r \mapsto \tilde{\lambda}(\tilde{\mathbf{x}}_r)$ ; substituting into (??) gives the *single*-argument residual  $\mathbf{r}(\tilde{\mathbf{x}}_r)$  that the corrector solves. The contact set is encoded by the time-domain Boolean mask

$$m_i = \mathbf{1}[\lambda_p(t_i) > 0] \in \{0, 1\}, \quad \mathbf{m} = \text{diag}(m_1, \dots, m_{N_t}). \quad (21)$$

## 4 The primary Newton unknown

**Claim.** *The algorithmic Newton unknown is the multiharmonic relative displacement  $\tilde{\mathbf{x}}_r$  (its  $2N_h n_{\text{int}}$  real DOF), not the contact force  $\tilde{\lambda}$ .*

*Justification from the residual/Jacobian structure (not convention):*

1. After (??)–(??),  $\tilde{\lambda}$  is an explicit *function*  $\tilde{\lambda}(\tilde{\mathbf{x}}_r)$ . The residual (??) therefore depends on  $\tilde{\mathbf{x}}_r$  *only*; the root-finding lives in  $\mathbb{R}^{2N_h n_{\text{int}}}$  and the Jacobian  $\partial \mathbf{r} / \partial \tilde{\mathbf{x}}_r$  (§??) is *square* in that space.
2. Promoting  $\tilde{\lambda}$  to an independent unknown would require appending the prediction equation and the complementarity (??) as extra equations, *doubling* the system and re-introducing the explicit Lagrange multipliers DLFT is designed to avoid (Vadcard §2.4.1; this is the very difference between DLFT and a wet/saddle-point Lagrangian, cf. van der Seijs Eq. 3.28). The admittance form makes this concrete:  $\mathbf{Y}_r$  in (??) already *is* the condensed interface operator, so no auxiliary force DOF are needed.
3. Consequently the continuation tangent and step are taken in  $(\tilde{\mathbf{x}}_r, \omega)$ . The contact force is never a state: it is recomputed from  $\tilde{\mathbf{x}}_r$  at *every* residual evaluation.

**Distinguish two notions.** (i) *Choice of unknowns (algorithmic):*  $\tilde{\mathbf{x}}_r$ , as argued above. (ii) *Initial guess at each continuation step (predictor):* the tangent predictor extrapolates the converged  $(\tilde{\mathbf{x}}_r, \omega)$  of the previous point; equivalently the linear relative response  $\tilde{\mathbf{x}}_r^{(0)} = \mathbf{f}_{\text{adm}}$  (no contact) is a natural cold start. **In both cases the carried quantity is the relative-displacement Fourier coefficients, never the contact forces** –  $\tilde{\lambda}$  is regenerated by (??)–(??).

## 5 Analytic Jacobian

The corrector and the arc-length predictor need  $\partial \mathbf{r} / \partial \tilde{\mathbf{x}}_r$  and  $\partial \mathbf{r} / \partial \omega$ . Both follow from (??) by the chain rule through the slaving map.

### 5.1 Derivative w.r.t. the relative-displacement coefficients

From (??),

$$\frac{\partial \mathbf{r}}{\partial \tilde{\mathbf{x}}_r} = \mathbf{I} + \mathbf{Y}_r \frac{\partial \tilde{\lambda}}{\partial \tilde{\mathbf{x}}_r}. \quad (22)$$

The slaving map (??)–(??) differentiates in three stages.

**(a) Prediction (frequency, holomorphic).** From (??),  $\tilde{\lambda}_p = (\mathbf{Z}_r \mathbf{f}_{\text{adm}} - \varepsilon \tilde{\mathbf{g}}_0) + (\varepsilon \mathbf{I} - \mathbf{Z}_r) \tilde{\mathbf{x}}_r$ , hence

$$\frac{\partial \tilde{\lambda}_p}{\partial \tilde{\mathbf{x}}_r} = \varepsilon \mathbf{I} - \mathbf{Z}_r \quad (\text{Vadcard A.3: } -\mathbf{Z}_r - \varepsilon \mathbf{I} \text{ in his sign convention}). \quad (23)$$

This block is *complex-linear* (holomorphic): in the real/imaginary (RI) split it has the structure  $\begin{bmatrix} \text{Re} & -\text{Im} \\ \text{Im} & \text{Re} \end{bmatrix}$ .

**(b) Correction (time, conditional / non-holomorphic).** The  $\max(0, \cdot)$  in (??) has derivative  $m_i$  pointwise (Vadcard A.4): keep the predicted sensitivity where there is contact, zero it where there is separation. With the mask (??),

$$\frac{\partial \lambda(t_i)}{\partial \tilde{\mathbf{x}}_r} = \begin{cases} \frac{\partial \lambda_p(t_i)}{\partial \tilde{\mathbf{x}}_r} & \lambda_p(t_i) > 0 \\ \mathbf{0} & \lambda_p(t_i) \leq 0 \end{cases} = \mathbf{m} \mathbf{\Gamma} \frac{\partial \tilde{\lambda}_p}{\partial \tilde{\mathbf{x}}_r}. \quad (24)$$

**(c) Re-projection (DFT).** Transform back with  $\mathbf{\Gamma}^+$ :

$$\boxed{\frac{\partial \tilde{\lambda}}{\partial \tilde{\mathbf{x}}_r} = \mathbf{\Gamma}^+ \mathbf{m} \mathbf{\Gamma} (\varepsilon \mathbf{I} - \mathbf{Z}_r)} \implies \frac{\partial \mathbf{r}}{\partial \tilde{\mathbf{x}}_r} = \mathbf{I} + \mathbf{Y}_r \mathbf{\Gamma}^+ \mathbf{m} \mathbf{\Gamma} (\varepsilon \mathbf{I} - \mathbf{Z}_r). \quad (25)$$

**Why this is *not* a holomorphic FRF map (the key correctness point).** The factors  $\mathbf{Y}_r$  and  $(\varepsilon \mathbf{I} - \mathbf{Z}_r)$  are complex-linear, but  $\mathbf{m}$  is a real diagonal acting *in the time domain* between  $\mathbf{\Gamma}$  and  $\mathbf{\Gamma}^+$ . The composite  $\mathbf{\Gamma}^+ \mathbf{m} \mathbf{\Gamma}$  is therefore *real-linear but not complex-linear*: in the RI split it is a *general* block  $\begin{bmatrix} \mathbf{A}_{RR} & \mathbf{A}_{RI} \\ \mathbf{A}_{IR} & \mathbf{A}_{II} \end{bmatrix}$  with  $\mathbf{A}_{RR} \neq \mathbf{A}_{II}$  and  $\mathbf{A}_{RI} \neq -\mathbf{A}_{IR}$  in general. Hence (??) **must** be assembled with the real operators  $\mathbf{\Gamma}, \mathbf{\Gamma}^+$  of (??); collapsing it into the holomorphic form  $\begin{bmatrix} \text{Re} & -\text{Im} \\ \text{Im} & \text{Re} \end{bmatrix}$  of the complex matrix  $\mathbf{I} + \mathbf{Y}_r(\varepsilon \mathbf{I} - \mathbf{Z}_r)$  is only valid in *persistent* contact ( $\mathbf{m} = \mathbf{I}$ , where  $\mathbf{\Gamma}^+ \mathbf{\Gamma} = \mathbf{I}$ ) and is wrong as soon as the contact set is partial. (This was the dominant bug in the original `compute_jacobian_of_residue_RI`.) When  $\mathbf{m} = \mathbf{I}$  the boxed result correctly collapses to  $\partial \mathbf{r} / \partial \tilde{\mathbf{x}}_r = \varepsilon \mathbf{Y}_r$ .

## 5.2 Derivative w.r.t. frequency

Here  $\tilde{\mathbf{x}}_r$  is held fixed and the  $\omega$ -dependence enters through  $\mathbf{Y}(\omega)$ . Per-harmonic,

$$\frac{\partial \mathbf{Z}_n}{\partial \omega} = -2n^2 \omega \mathbf{M} + i n \mathbf{C}, \quad \frac{\partial \mathbf{Y}_n}{\partial \omega} = -\mathbf{Y}_n \frac{\partial \mathbf{Z}_n}{\partial \omega} \mathbf{Y}_n, \quad (26)$$

and  $\partial \mathbf{Y}_r / \partial \omega = \mathbf{B}_c (\partial \mathbf{Y} / \partial \omega) \mathbf{B}_e^\top$ ,  $\partial \mathbf{f}_{\text{adm}} / \partial \omega = \mathbf{B}_c (\partial \mathbf{Y} / \partial \omega) \tilde{\mathbf{f}}_{\text{ext}}$ ,  $\partial \mathbf{Z}_r / \partial \omega = -\mathbf{Z}_r (\partial \mathbf{Y}_r / \partial \omega) \mathbf{Z}_r$ . Differentiating (??),

$$\frac{\partial \mathbf{r}}{\partial \omega} = \frac{\partial \mathbf{Y}_r}{\partial \omega} \tilde{\lambda} + \mathbf{Y}_r \frac{\partial \tilde{\lambda}}{\partial \omega} - \frac{\partial \mathbf{f}_{\text{adm}}}{\partial \omega}. \quad (27)$$

The contact-force term uses the *same* time-domain mask as (??); since  $\partial \tilde{\lambda}_p / \partial \omega$  is a single (real-signal) vector its IDFT is unambiguous:

$$\frac{\partial \tilde{\lambda}}{\partial \omega} = \mathbf{\Gamma}^+ \mathbf{m} \mathbf{\Gamma} \underbrace{\left[ \frac{\partial \mathbf{Z}_r}{\partial \omega} (\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r) + \mathbf{Z}_r \frac{\partial \mathbf{f}_{\text{adm}}}{\partial \omega} \right]}_{\partial \tilde{\lambda}_p / \partial \omega}, \quad \frac{\partial \tilde{\lambda}_p}{\partial \omega} = \mathbf{Z}_r \frac{\partial (\mathbf{B} \mathbf{Y})}{\partial \omega} (\tilde{\mathbf{f}}_{\text{ext}} - \mathbf{B}_e^\top \mathbf{Z}_r (\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r)). \quad (28)$$

The first two terms of (??) combine to  $\mathbf{B}_c (\partial \mathbf{Y} / \partial \omega) (\mathbf{B}_e^\top \tilde{\lambda} - \tilde{\mathbf{f}}_{\text{ext}})$ , matching the implemented `compute_derivative_wrt_omega_RI` (verified against finite differences to  $\sim 10^{-13}$ ).

## 6 Properties, conditioning and algorithm

### 6.1 $\varepsilon$ -independence of the solution

At convergence ( $\mathbf{r} = \mathbf{0}$ ) the solution is independent of  $\varepsilon$ . With  $\lambda = \lambda_p + \lambda_c$  and  $\lambda_c = \max(0, -\lambda_p) \geq 0$ , substituting the prediction (??) into (??) gives  $\mathbf{r} = \mathbf{Y}_r(\tilde{\lambda}_c - \varepsilon \tilde{\mathbf{g}})$ , i.e. the impedance residual  $\mathbf{Z}_r \mathbf{r} = \tilde{\lambda}_c - \varepsilon \tilde{\mathbf{g}}$ . The correction satisfies  $0 \leq \frac{1}{\varepsilon} \lambda_c = g_p \perp \lambda \geq 0$  (Vadcard Eqs. 44–45), so the root  $\tilde{\mathbf{g}}_p = \tilde{\mathbf{g}}$  is reached independently of  $\varepsilon$ . Two practical caveats: (i)  $\varepsilon$  must be *large enough* to detect contact through the penalty – in the test problem (stiffness  $\sim 1$ ) convergence requires  $\varepsilon \gtrsim 10^6$  and is robust at  $10^8$  (Vadcard sweeps  $\varepsilon \in [10^8, 2 \cdot 10^{12}]$ ); (ii) larger  $\varepsilon$  amplifies the harmonic-truncation oscillations of the non-smooth  $\lambda(t)$ , so the *absolute* residual at the floor scales with  $\varepsilon$  – the convergence test must be relative (below).

### 6.2 Conditioning: why admittance vs. impedance matters

The two residuals are related by  $\mathbf{r}_{\text{adm}} = \mathbf{Y}_r \mathcal{H}_{\text{imp}}$ . Near a substructure resonance  $\mathbf{Y}_r$  is large and ill-conditioned, so admittance-form Newton is more sensitive there; this is intrinsic to FBS with measured/numeric FRFs (van der Seijs after Eq. 3.26). The admittance form is nevertheless required for experimental FRFs and is what the FBS solver uses; the cure is a relative/scaled convergence criterion and arc-length continuation with adaptive steps.

### 6.3 Solver criterion

Because of the truncation floor, the corrector accepts  $\tilde{\mathbf{x}}_r$  when *either* the residual is small relative to its value at the predictor,  $\|\mathbf{r}\| \leq \eta_{\text{rel}} \|\mathbf{r}^{(0)}\|$ , *or* the Newton step stagnates,  $\|\Delta \tilde{\mathbf{x}}_r\| \leq \eta_{\text{stag}} \|\tilde{\mathbf{x}}_r\|$  (the floor is reached), in addition to the original absolute test  $\|\mathbf{r}\| < \eta_{\text{abs}}$ .

### 6.4 Algorithm (drop-in replacement for AFT)

**DLFT residual/Jacobian at  $(\tilde{\mathbf{x}}_r, \omega)$**  – same interface as AFT:

1. Build  $\mathbf{Y}(\omega)$  (numeric from  $\mathbf{M}, \mathbf{C}, \mathbf{K}$ , or interpolated FRF);  $\mathbf{Y}_r = \mathbf{B}_c \mathbf{Y} \mathbf{B}_e^\top$ ,  $\mathbf{f}_{\text{adm}} = \mathbf{B}_c \mathbf{Y} \tilde{\mathbf{f}}_{\text{ext}}$ ,  $\mathbf{Z}_r = \mathbf{Y}_r^{-1}$ .
2. *Prediction*:  $\tilde{\lambda}_p = \mathbf{Z}_r(\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r) + \varepsilon(\tilde{\mathbf{x}}_r - \tilde{\mathbf{g}}_0)$ ;  $\lambda_p = \Gamma \tilde{\lambda}_p$  (IDFT).
3. *Correction*:  $\lambda = \max(0, \lambda_p)$ , mask  $\mathbf{m}$ ;  $\tilde{\lambda} = \Gamma^+ \lambda$  (DFT).
4. *Residual*:  $\mathbf{r} = \tilde{\mathbf{x}}_r + \mathbf{Y}_r \tilde{\lambda} - \mathbf{f}_{\text{adm}}$ .
5. *Jacobian*:  $\partial \mathbf{r} / \partial \tilde{\mathbf{x}}_r = \mathbf{I} + \mathbf{Y}_r \Gamma^+ \mathbf{m} \Gamma (\varepsilon \mathbf{I} - \mathbf{Z}_r)$ ;  $\partial \mathbf{r} / \partial \omega$  from (??)–(??).

The corrector solves  $\mathbf{r}(\tilde{\mathbf{x}}_r) = \mathbf{0}$  for  $\tilde{\mathbf{x}}_r$  inside the predictor–corrector arc-length continuation in  $\omega$ ; *only* steps 2–3 (AFT  $\mathbf{f}_{\text{nl}}$ ) and step 5 differ between AFT and DLFT. Response recovery uses  $\hat{\mathbf{u}} = \mathbf{Y}(\tilde{\mathbf{f}}_{\text{ext}} - \mathbf{B}_e^\top \tilde{\lambda})$  (Soleimani Eqs. 25–26).

## 7 Map to the pyhbm implementation

| Symbol                                                                          | Meaning                                   | FBS_DLFT member                              |
|---------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------|
| $\tilde{\mathbf{x}}_r$                                                          | rel. displacement coeff. (Newton unknown) | <code>x.fourier.coefficients</code>          |
| $\mathbf{B}_c = \mathbf{B}_e = \mathbf{B}$                                      | signed Boolean (multiharmonic)            | <code>self.B_fourier</code>                  |
| $\mathbf{Y}$                                                                    | block-diag substructure admittance        | <code>compute_FRF(omega)</code>              |
| $\mathbf{Y}_r = \mathbf{B}\mathbf{Y}\mathbf{B}^\top$                            | interface admittance                      | <code>_get_Yr</code>                         |
| $\mathbf{f}_{\text{adm}} = \mathbf{B}\mathbf{Y}\tilde{\mathbf{f}}_{\text{ext}}$ | free relative response                    | <code>_get_Fext_admr</code>                  |
| $\mathbf{Z}_r(\mathbf{f}_{\text{adm}} - \tilde{\mathbf{x}}_r)$                  | balancing force (pre-penalty)             | <code>_get_Zr_rhs</code>                     |
| $\tilde{\boldsymbol{\lambda}} = \boldsymbol{\Gamma}^+ \max(0, \lambda_p)$       | corrected contact force                   | <code>_get_lambda_corrected</code>           |
| $\mathbf{m}$                                                                    | contact mask $\mathbf{1}[\lambda_p > 0]$  | <code>x.contact_mask</code>                  |
| $\boldsymbol{\Gamma}, \boldsymbol{\Gamma}^+$                                    | real IDFT/DFT (??)                        | <code>self._idft_RI, self._dft_RI</code>     |
| $\partial \mathbf{r} / \partial \tilde{\mathbf{x}}_r$                           | Eq. (??)                                  | <code>compute_jacobian_of_residue_RI</code>  |
| $\partial \mathbf{r} / \partial \omega$                                         | Eq. (??)                                  | <code>compute_derivative_wrt_omega_RI</code> |

**Diagnosed defects of the original code (corrected per this note).** (b) the matrix  $\partial \tilde{\boldsymbol{\lambda}} / \partial \tilde{\mathbf{x}}_r$  was built with `rfft/irfft` on *complex* columns and then wrapped by the holomorphic `FRF_to_RI`, i.e. (??) assembled as if  $\boldsymbol{\Gamma}^+ \mathbf{m} \boldsymbol{\Gamma}$  were complex-linear – wrong for partial contact; (c)  $\varepsilon = 10^2$  was orders of magnitude too small; (a) the sign of the penalty is self-consistent with a wall at  $+g_0$  (equivalent to Vadcard up to the sign of  $x_r$ ); (d) the time-domain correction (??) before the DFT was already correct;  $\partial \mathbf{r} / \partial \omega$  was already correct.

## References

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- [3] H. Soleimani, N. Aage, *A closed-form nonlinear dynamic substructuring method*, CMAME **439** (2025) 117882 (Eq. 24).
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